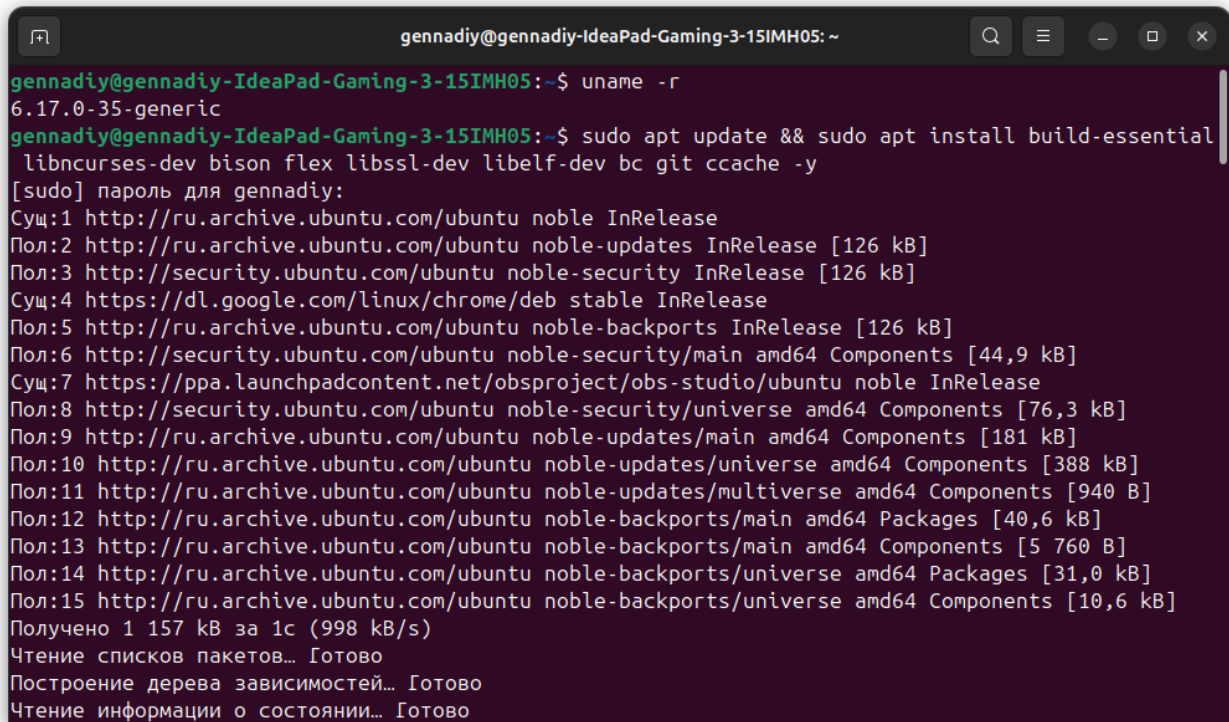


Отчет к заданию №17 — по ядру Linux

1) Установка инструментов сборки и зависимостей `sudo apt update && sudo apt install build-essential libncurses-dev bison flex libssl-dev libelf-dev bc git ccache -y`



```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ uname -r  
6.17.0-35-generic  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ sudo apt update && sudo apt install build-essential  
libncurses-dev bison flex libssl-dev libelf-dev bc git ccache -y  
[sudo] пароль для gennadiy:  
Сущ:1 http://ru.archive.ubuntu.com/ubuntu noble InRelease  
Пол:2 http://ru.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Пол:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Сущ:4 https://dl.google.com/linux/chrome/deb stable InRelease  
Пол:5 http://ru.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]  
Пол:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [44,9 kB]  
Сущ:7 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease  
Пол:8 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76,3 kB]  
Пол:9 http://ru.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]  
Пол:10 http://ru.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]  
Пол:11 http://ru.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]  
Пол:12 http://ru.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40,6 kB]  
Пол:13 http://ru.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5 760 B]  
Пол:14 http://ru.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [31,0 kB]  
Пол:15 http://ru.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10,6 kB]  
Получено 1 157 kB за 1с (998 kB/s)  
Чтение списков пакетов... Готово  
Построение дерева зависимостей... Готово  
Чтение информации о состоянии... Готово
```

Рисунок 1 — установка инструментов сборки и зависимостей

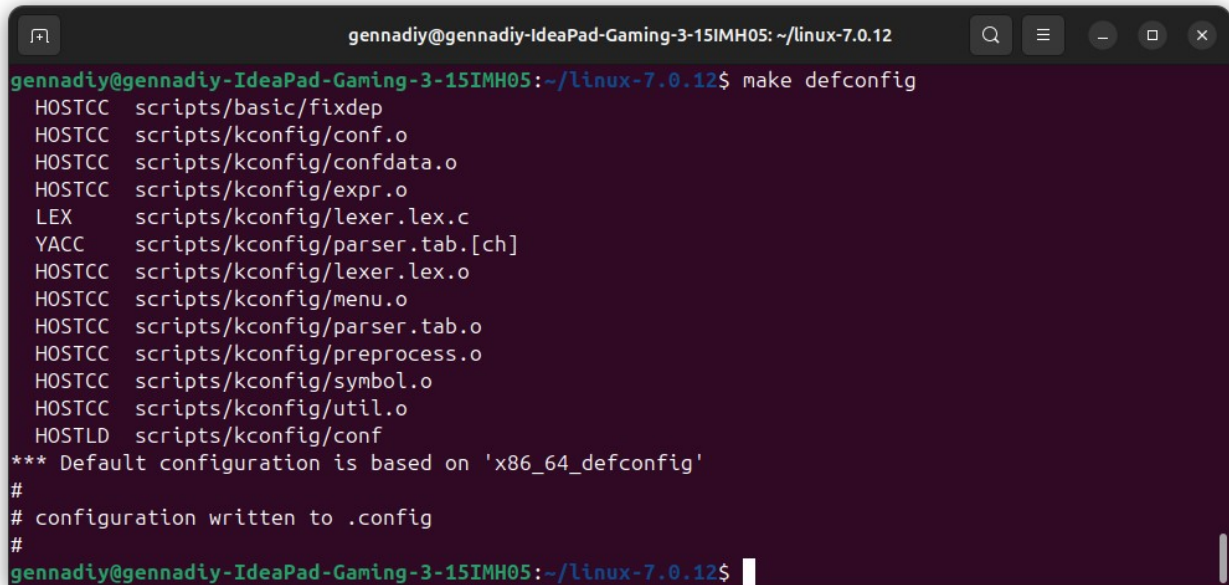
2) Скачивание и распаковка исходных кодов ядра
`wget https://cdn.kernel.org/pub/linux/kernel/v7.x/linux-7.0.12.tar.xz`
`tar -xf linux-7.0.12.tar.xz`
`cd linux-7.0.12/`



```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/linux-7.0.12  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ wget https://cdn.kernel.org/pub/linux/kernel/v7.x/linux-7.0.12.tar.xz  
--2026-06-15 01:54:32-- https://cdn.kernel.org/pub/linux/kernel/v7.x/linux-7.0.12.tar.xz  
Распознаётся cdn.kernel.org (cdn.kernel.org)... 146.75.117.176, 2a04:4e42:8d::432  
Подключение к cdn.kernel.org (cdn.kernel.org)[146.75.117.176]:443... соединение установлено.  
HTTP-запрос отправлен. Ожидание ответа... 200 OK  
Длина: 157240476 (150M) [application/x-xz]  
Сохранение в: 'linux-7.0.12.tar.xz'  
  
linux-7.0.12.tar.xz 100%[=====] 149,96M 189KB/s за 9м 25с  
  
2026-06-15 02:03:58 (272 KB/s) - 'linux-7.0.12.tar.xz' сохранён [157240476/157240476]  
  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ tar -xf linux-7.0.12.tar.xz  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ ls  
Команда «ls» не найдена, но может быть установлена с помощью:  
sudo apt install mono-devel  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ ls  
index.html.1 linux-7.0.12 snap Документы Изображения Общедоступные Шаблоны  
input.txt linux-7.0.12.tar.xz Видео Загрузки Музыка 'Рабочий стол'  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ cd linux-7.0.12/  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/linux-7.0.12$
```

Рисунок 2 — скачивание и распаковка исходных кодов ядра

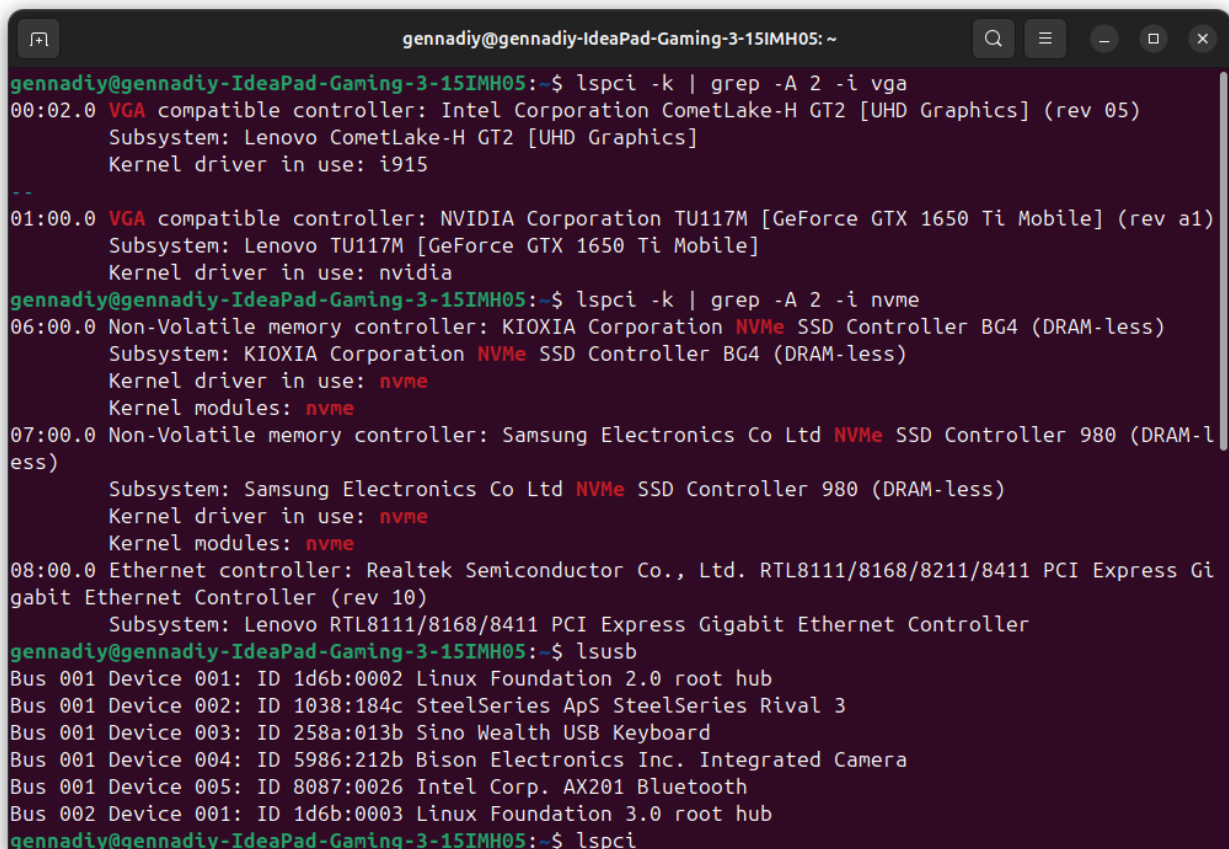
3) Инициализация стандартной конфигурации по умолчанию make defconfig



```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/linux-7.0.12
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/linux-7.0.12$ make defconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/kconfig/conf.o
HOSTCC scripts/kconfig/confdata.o
HOSTCC scripts/kconfig/expr.o
LEX scripts/kconfig/lexer.lex.c
YACC scripts/kconfig/parser.tab.[ch]
HOSTCC scripts/kconfig/lexer.lex.o
HOSTCC scripts/kconfig/menu.o
HOSTCC scripts/kconfig/parser.tab.o
HOSTCC scripts/kconfig/preprocess.o
HOSTCC scripts/kconfig/symbol.o
HOSTCC scripts/kconfig/util.o
HOSTLD scripts/kconfig/conf
*** Default configuration is based on 'x86_64_defconfig'
#
# configuration written to .config
#
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/linux-7.0.12$
```

Рисунок 3 — скачивание и распаковка исходных кодов ядра

4) Анализ оборудования через lspci lsusb



```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ lspci -k | grep -A 2 -i vga
00:02.0 VGA compatible controller: Intel Corporation CometLake-H GT2 [UHD Graphics] (rev 05)
Subsystem: Lenovo CometLake-H GT2 [UHD Graphics]
Kernel driver in use: i915
--
01:00.0 VGA compatible controller: NVIDIA Corporation TU117M [GeForce GTX 1650 Ti Mobile] (rev a1)
Subsystem: Lenovo TU117M [GeForce GTX 1650 Ti Mobile]
Kernel driver in use: nvidia
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ lspci -k | grep -A 2 -i nvme
06:00.0 Non-Volatile memory controller: KIOXIA Corporation NVMe SSD Controller BG4 (DRAM-less)
Subsystem: KIOXIA Corporation NVMe SSD Controller BG4 (DRAM-less)
Kernel driver in use: nvme
Kernel modules: nvme
07:00.0 Non-Volatile memory controller: Samsung Electronics Co Ltd NVMe SSD Controller 980 (DRAM-less)
Subsystem: Samsung Electronics Co Ltd NVMe SSD Controller 980 (DRAM-less)
Kernel driver in use: nvme
Kernel modules: nvme
08:00.0 Ethernet controller: Realtek Semiconductor Co., Ltd. RTL8111/8168/8211/8411 PCI Express Gigabit Ethernet Controller (rev 10)
Subsystem: Lenovo RTL8111/8168/8411 PCI Express Gigabit Ethernet Controller
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ lsusb
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 002: ID 1038:184c SteelSeries ApS SteelSeries Rival 3
Bus 001 Device 003: ID 258a:013b Sino Wealth USB Keyboard
Bus 001 Device 004: ID 5986:212b Bison Electronics Inc. Integrated Camera
Bus 001 Device 005: ID 8087:0026 Intel Corp. AX201 Bluetooth
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ lspci
```

Рисунок 4 — анализ оборудования

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~  
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ lspci  
00:00.0 Host bridge: Intel Corporation 10th Gen Core Processor Host Bridge/DRAM Registers (rev 02)  
00:01.0 PCI bridge: Intel Corporation 6th-10th Gen Core Processor PCIe Controller (x16) (rev 02)  
00:02.0 VGA compatible controller: Intel Corporation CometLake-H GT2 [UHD Graphics] (rev 05)  
00:04.0 Signal processing controller: Intel Corporation Xeon E3-1200 v5/E3-1500 v5/6th Gen Core Processor Thermal Subsystem (rev 02)  
00:08.0 System peripheral: Intel Corporation Xeon E3-1200 v5/v6 / E3-1500 v5 / 6th/7th/8th Gen Core Processor Gaussian Mixture Model  
00:12.0 Signal processing controller: Intel Corporation Comet Lake PCH Thermal Controller  
00:14.0 USB controller: Intel Corporation Comet Lake USB 3.1 xHCI Host Controller  
00:14.2 RAM memory: Intel Corporation Comet Lake PCH Shared SRAM  
00:14.3 Network controller: Intel Corporation Comet Lake PCH CNVi WiFi  
00:15.0 Serial bus controller: Intel Corporation Comet Lake PCH Serial IO I2C Controller #0  
00:15.1 Serial bus controller: Intel Corporation Comet Lake PCH Serial IO I2C Controller #1  
00:16.0 Communication controller: Intel Corporation Comet Lake HECI Controller  
00:17.0 SATA controller: Intel Corporation Device 06d3  
00:1b.0 PCI bridge: Intel Corporation Comet Lake PCI Express Root Port #17 (rev f0)  
00:1d.0 PCI bridge: Intel Corporation Comet Lake PCI Express Root Port #9 (rev f0)  
00:1d.6 PCI bridge: Intel Corporation Device 06b6 (rev f0)  
00:1f.0 ISA bridge: Intel Corporation Comet Lake LPC Controller  
00:1f.4 Audio device: Intel Corporation Comet Lake PCH cAVS  
00:1f.5 SMBus: Intel Corporation Comet Lake PCH SMBus Controller  
01:00.0 VGA compatible controller: NVIDIA Corporation TU117M [GeForce GTX 1650 Ti Mobile] (rev a1)  
01:00.1 Audio device: NVIDIA Corporation Device 10fa (rev a1)  
06:00.0 Non-Volatile memory controller: KIOXIA Corporation NVMe SSD Controller BG4 (DRAM-less)  
07:00.0 Non-Volatile memory controller: Samsung Electronics Co Ltd NVMe SSD Controller 980 (DRAM-less)  
08:00.0 Ethernet controller: Realtek Semiconductor Co., Ltd. RTL8111/8168/8211/8411 PCI Express Gigabit Ethernet Controller (rev 10)  
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$
```

Рисунок 5 — анализ оборудования

5) Вводим команду `make menuconfig` и выполняем следующие настройки параметров ядра Linux:

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/linux-7.0.12  
#config - Linux/x86 7.0.12 Kernel Configuration  
-> Search (DRM_I915) -> Direct Rendering Manager (XFree86 4.1.0 and higher DRI support)  
    Direct Rendering Manager (XFree86 4.1.0 and higher DRI support)  
    Arrow keys navigate the menu. <Enter> selects submenus ---> (or empty submenus ----).  
    Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes, <M> modularizes features.  
    Press <Esc><Esc> to exit, <?> for Help, </> for Search. Legend: [*] built-in [ ] excluded  
    <M> module < > module capable  
  
    --- Direct Rendering Manager (XFree86 4.1.0 and higher DRI support)  
        DRM debugging options --->  
        [ ] Display a user-friendly message when a kernel panic occurs  
        Supported DRM clients --->  
        [ ] Allow to specify an EDID data set instead of probing for it  
        [ ] Enable DisplayPort CEC-Tunneling-over-AUX HDMI support  
        [ ] DRM DP AUX Interface  
        < > AMD GPU  
        ARM devices ----  
        < > AST server chips  
        Display Interface Bridges --->  
        < > ETNAVIV (DRM support for Vivante GPU IP cores)  
        < > Intel GMA500/600/3600/3650 KMS Framebuffer  
        < > GUD USB Display  
        < > DRM Support for Hisilicon Hibmc  
        [*] Intel 8xx/9xx/G3x/G4x/HD Graphics  
            ( ) Force probe i915 for selected Intel hardware IDs  
            [*] Enable capturing GPU state following a hang  
            [*] Compress GPU error state  
            [*] Always enable userptr support  
        < > Matrox G200  
        < > Nouveau (NVIDIA) cards  
        ↓(+)  
  
    <select> < Exit > < Help > < Save > < Load >
```

Рисунок 6 — включение встроенной видеокарты

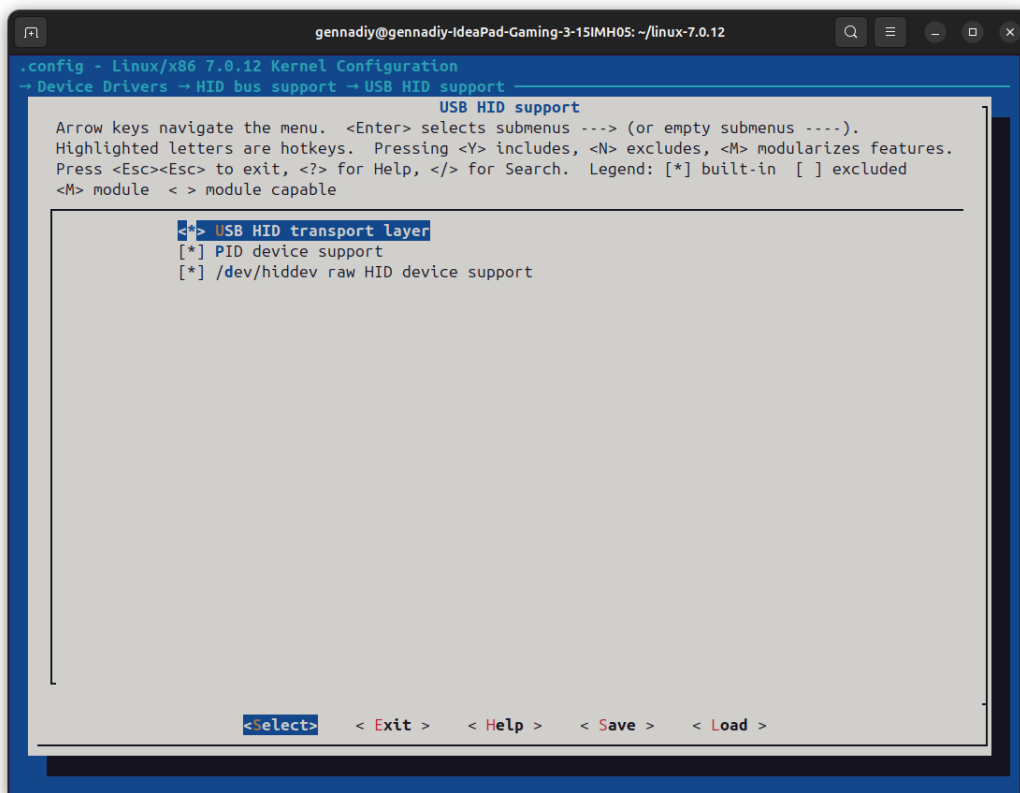


Рисунок 8 — настройка поддержки USB-устройств

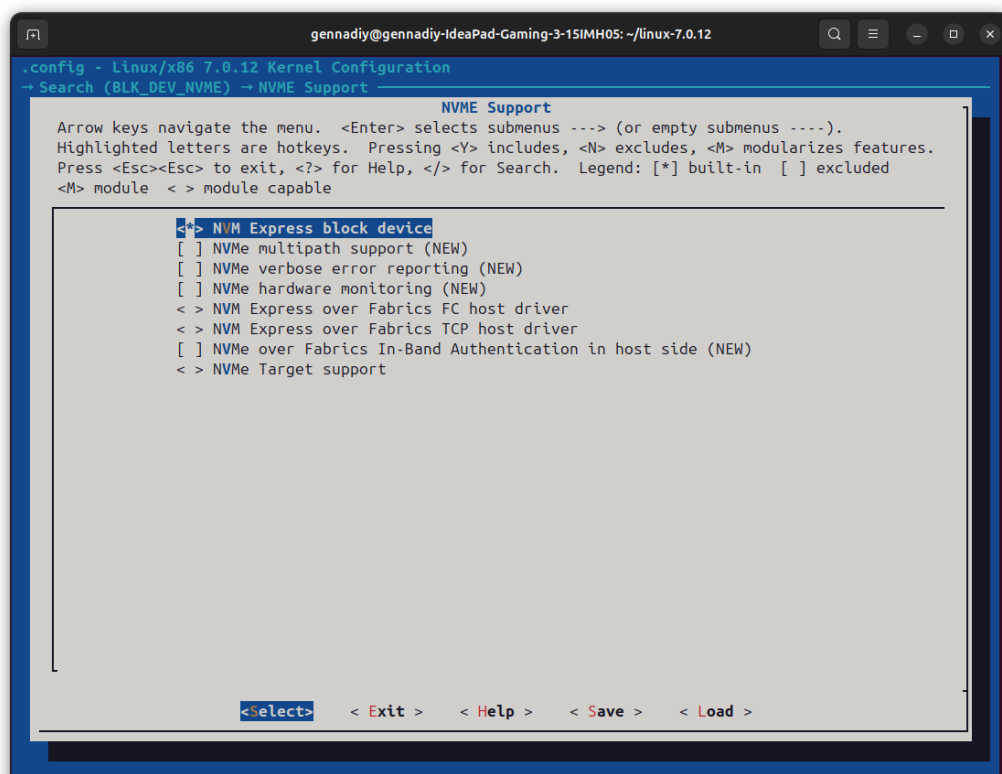


Рисунок 9 — конфигурация драйвера ssd — накопителя

6) Запуск сборки ядра командой (с компиляцией в многопоточном режиме) `make -j8 bzImage && make -j8 modules && sudo make modules_install && sudo make install`

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/linux-7.0.12
[sudo] пароль для gennadiy:
SYMLINK /lib/modules/7.0.12/build
INSTALL /lib/modules/7.0.12/modules.order
INSTALL /lib/modules/7.0.12/modules.builtin
INSTALL /lib/modules/7.0.12/modules.builtin.modinfo
INSTALL /lib/modules/7.0.12/kernel/fs/efivarfs/efivarfs.ko
INSTALL /lib/modules/7.0.12/kernel/drivers/thermal/intel/x86_pkg_temp_thermal.ko
INSTALL /lib/modules/7.0.12/kernel/net/netfilter/nf_log_syslog.ko
INSTALL /lib/modules/7.0.12/kernel/net/netfilter/xt_mark.ko
INSTALL /lib/modules/7.0.12/kernel/net/netfilter/xt_LOG.ko
INSTALL /lib/modules/7.0.12/kernel/net/netfilter/xt_MASQUERADE.ko
INSTALL /lib/modules/7.0.12/kernel/net/netfilter/xt_addrtype.ko
INSTALL /lib/modules/7.0.12/kernel/net/ipv4/netfilter/nf_reject_ipv4.ko
INSTALL /lib/modules/7.0.12/kernel/net/ipv6/netfilter/nf_reject_ipv6.ko
DEPMOD /lib/modules/7.0.12
INSTALL /boot
run-parts: executing /etc/kernel/postinst.d/dkms 7.0.12 /boot/vmlinuz-7.0.12
* dkms: running auto installation service for kernel 7.0.12
The kernel is be built without module signing facility, modules won't be signed

Building module:
Cleaning build area...(bad exit status: 2)
unset ARCH; [ ! -h /usr/bin/cc ] && export CC=/usr/bin/gcc; env NV_VERBOSE=1 'make' -j8 NV_EXCLUDE_BUILD_MOD
ULES='' KERNEL_UNAME=7.0.12 IGNORE_XEN_PRESENCE=1 IGNORE_CC_MISMATCH=1 SYSSRC=/lib/modules/7.0.12/build LD=/
usr/bin/ld.bfd CONFIG_X86_KERNEL_IBT= modules...(bad exit status: 2)
ERROR (dkms apport): kernel package linux-headers-7.0.12 is not supported
Error! Bad return status for module build on kernel: 7.0.12 (x86_64)
Consult /var/lib/dkms/nvidia/595.71.05/build/make.log for more information.
dkms autoinstall on 7.0.12/x86_64 failed for nvidia(10)
Error! One or more modules failed to install during autoinstall.
Refer to previous errors for more information.
* dkms: autoinstall for kernel 7.0.12 [fail]
run-parts: /etc/kernel/postinst.d/dkms exited with return code 11
make[1]: *** [arch/x86/Makefile:319: install] Ошибка 11
make: *** [Makefile:248: __sub-make] Ошибка 2
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/linux-7.0.12$
```

Рисунок 10 — результаты сборки ядра

Сборка ядра с базовыми драйверами завершена успешно. Статус fail относится к утилите dkms, которая не смогла собрать драйвер дискретной видеокарты nvidia. Для решения этой проблемы отключаем скрипт dkms и запускаем повторную установку с обновлением конфигурации GRUB:

- Отключение скрипта dkms: `sudo chmod -x /etc/kernel/postinst.d/dkms`
- Запуск повторной установки: `sudo make install`
- Возврат прав скрипту dkms: `sudo chmod +x /etc/kernel/postinst.d/dkms`
- Обновление конфигурации GRUB: `sudo update-grub`

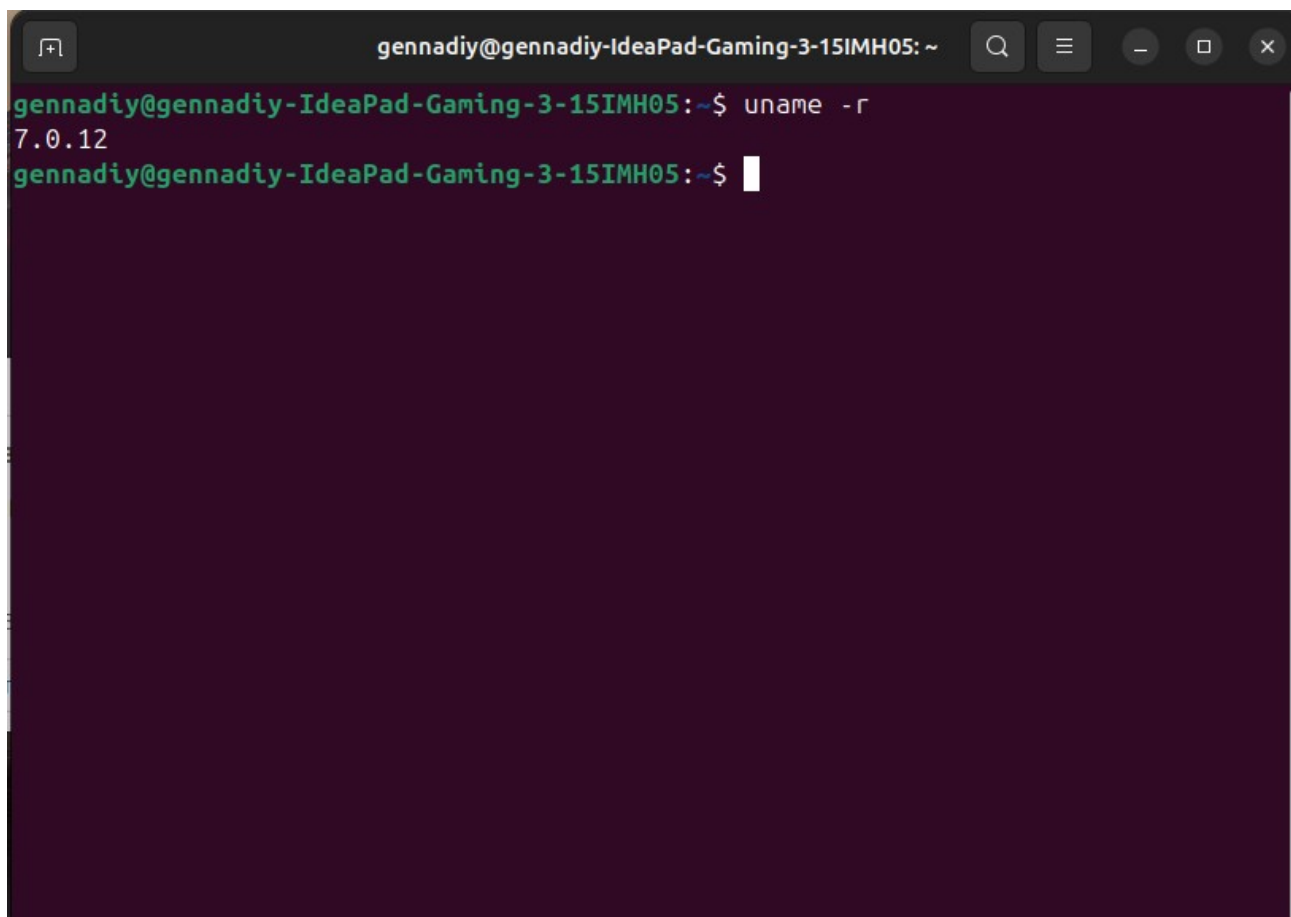

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~/linux-7.0.12
Found initrd image: /boot/initrd.img-6.17.0-29-generic
Found memtest86+ 64bit EFI image: /boot/memtest86+x64.efi
Warning: os-prober will be executed to detect other bootable partitions.
Its output will be used to detect bootable binaries on them and create new boot entries.
Found Windows Boot Manager on /dev/nvme1n1p2@/efi/Microsoft/Boot/bootmgfw.efi
Adding boot menu entry for UEFI Firmware Settings ...
done
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/linux-7.0.12$ sudo chmod +x /etc/kernel/postinst.d/dkms
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/linux-7.0.12$ sudo update-grub
Sourcing file '/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-7.0.12
Found initrd image: /boot/initrd.img-7.0.12
Found linux image: /boot/vmlinuz-7.0.12.old
Found initrd image: /boot/initrd.img-7.0.12
Found linux image: /boot/vmlinuz-6.17.0-35-generic
Found initrd image: /boot/initrd.img-6.17.0-35-generic
Found linux image: /boot/vmlinuz-6.17.0-29-generic
Found initrd image: /boot/initrd.img-6.17.0-29-generic
Found memtest86+ 64bit EFI image: /boot/memtest86+x64.efi
Warning: os-prober will be executed to detect other bootable partitions.
Its output will be used to detect bootable binaries on them and create new boot entries.
Found Windows Boot Manager on /dev/nvme1n1p2@/efi/Microsoft/Boot/bootmgfw.efi
Adding boot menu entry for UEFI Firmware Settings ...
done
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~/linux-7.0.12$
```

Рисунок 11 — обновление конфигурации grub

Через меню Advanced options загрузчика grub для запуска выбрана собранная версия ядра 7.0.12 (рисунок 12). После загрузки проверил версию ядра командой uname -r (рисунок 13)

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05: ~
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ sudo awk -F'|' '{print $2}' /boot/grub/grub.cfg
Ubuntu
Ubuntu, with Linux 7.0.12
Ubuntu, with Linux 7.0.12 (recovery mode)
Ubuntu, with Linux 7.0.12.old
Ubuntu, with Linux 7.0.12.old (recovery mode)
Ubuntu, with Linux 6.17.0-35-generic
Ubuntu, with Linux 6.17.0-35-generic (recovery mode)
Ubuntu, with Linux 6.17.0-29-generic
Ubuntu, with Linux 6.17.0-29-generic (recovery mode)
memtest86+
Memory test (memtest86+x64.efi, serial console)
Windows Boot Manager (on /dev/nvme1n1p2)
UEFI Firmware Settings
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$
```

Рисунок 12 — список доступных конфигураций ядер linux

A terminal window with a dark background. The title bar at the top shows the user 'gennadiy' and the host 'gennadiy-IdeaPad-Gaming-3-15IMH05'. The terminal displays a green prompt 'gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~\$' followed by the command 'uname -r'. The output '7.0.12' is shown on the next line. A second green prompt 'gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~\$' is visible with a white cursor.

```
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$ uname -r
7.0.12
gennadiy@gennadiy-IdeaPad-Gaming-3-15IMH05:~$
```

Рисунок 13 — проверка версии ядра после запуска